Contact

kelsoncm@gmail.com

www.linkedin.com/in/kelsoncm (LinkedIn)
github.com/kelsoncm (Other)
lattes.cnpq.br/1734494254835148 (Other)
orcid.org/0000-0001-6749-4656 (Other)

Top Skills

Secure Design Principles
Secure Software Development
Handling Vulnerability Reports

Languages

English (Professional Working)
Portuguese (Native or Bilingual)
Spanish (Elementary)

Certifications

Certificado de Atividade - VI Seminário Internacional de Educação a Distância - SEMEAD 2024
LFD121: Developing Secure Software
Certificado da Comissão Organizadora - VI Seminário Internacional de Educação a Distância - SEMEAD 2024
SKF100: Understanding the OWASP® Top 10 Security Threats
LFD125: Security for Software Development Managers

Kelson da Costa Medeiros

Senior Specialist in Educational Technologies | Researcher in Digital Innovation | Architect of Scalable Solutions for Health, Education & Governance
Natal, Rio Grande do Norte, Brazil

Summary

EN: Senior DevOps Specialist & Software Architect with 30+ years driving digital transformation across government, healthcare, and education sectors in Brazil. Currently leading AWS cloud infrastructure and CI/CD automation for Brazil's Ministry of Culture streaming platform while coordinating educational technology systems at IFRN. I specialize in architecting scalable solutions—from legacy system modernization (Delphi, Java/Spring, Oracle) to cloud-native platforms (Kubernetes, Terraform, FastAPI, Django)—with deep expertise in learning management systems (Moodle), healthcare ecosystems (AVASUS, LUES, PNTD serving millions of SUS workers), and enterprise document management. My technical leadership spans full-stack development (Python, PHP, Java, JavaScript), DevOps practices (Docker, GitLab CI/CD, AWS), database architecture (PostgreSQL, MySQL, Oracle), and Agile team management. I bridge technical innovation with institutional impact, delivering mission-critical systems that modernize public services and democratize access to education and healthcare across Brazil's 27 states.BR: Especialista Sênior em DevOps e Arquiteto de Software com mais de 30 anos de experiência impulsionando a transformação digital nos setores público, de saúde e educação no Brasil. Atualmente, lidero a infraestrutura de nuvem AWS e a automação de CI/CD para a plataforma de streaming do Ministério da Cultura, além de coordenar sistemas de tecnologia educacional no IFRN. Sou especialista em arquitetura de soluções escaláveis — desde a modernização de sistemas legados (Delphi, Java/Spring, Oracle) até plataformas nativas da nuvem (Kubernetes, Terraform, FastAPI, Django) — com profundo conhecimento em sistemas de gestão de aprendizagem (Moodle), ecossistemas de saúde (AVASUS, LUES, PNTD, atendendo milhões de trabalhadores do SUS) e gestão documental corporativa. Minha liderança técnica abrange desenvolvimento full-stack (Python, PHP, Java, JavaScript), práticas de DevOps (Docker, GitLab CI/CD, AWS), arquitetura

de banco de dados (PostgreSQL, MySQL, Oracle) e gestão de equipes ágeis. Conecto inovação técnica com impacto institucional, entregando sistemas de missão crítica que modernizam serviços públicos e democratizam o acesso à educação e à saúde nos 27 estados do Brasil.

Experience

Instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Norte

12 years 7 months

Educational Technology Coordinator

May 2017 - Present (9 years 3 months)
Natal, Rio Grande do Norte, Brazil

1. Management of teams of developers, designers, user support, equipment maintenance and networks.

Systems Analyst

January 2014 - May 2017 (3 years 5 months)
Natal, Rio Grande do Norte, Brazil

1. PHP development: project management of the new EaD portal and plugins and theme for Moodle; development of Moodle plugin for integration with the SUAP academic system.
2. PostgreSQL 9.x DBMS Administration: Definition of the redundancy project.
3. Governance: Beginning of the PDTI project.
4. Python and Django development: Analysis of management modules for research and innovation projects and project management for extension of SUAP (Unified Public Administration System) and development of a student selection process system. Project coordination of the new MOOC solution design on the IFRN TIMTec platform.

Núcleo de Excelência em Tecnologias Sociais - NEES

DevOps Specialist

April 2025 - Present (1 year 4 months)
Maceió, AL

DevOps Specialist leading the architecture, infrastructure as code, and deployment of the Ministry of Culture’s (MinC) AWS-based streaming platform, implementing GitLab CI/CD pipelines, Kubernetes and Helm-based orchestration, Terraform-managed IaaS, and CDN-backed content delivery to ensure high availability, scalability, and automation.

Laboratório de Inovação tecnológica em Saúde (LAIS/HUOL/UFRN)
IT Researcher
July 2015 - December 2025 (10 years 6 months)
Natal, RN

Software Development Manager at Campus Virtual, I lead the development of AVASUS (SUS e-Learning platform training workers, managers, and healthcare providers across Brazil's Unified Health System), LUES Ecosystem (nationwide syphilis prevention and control initiative), and PNTD (multi-state telediagnosis ecosystem serving 9 Brazilian states). My technical expertise spans backend development with Python/Django/FastAPI, PHP/Laravel Lumen, Celery task queues, RESTful APIs with Swagger documentation; frontend technologies including JavaScript/Vue.js/CSS; database management with PostgreSQL/SQL/Elasticsearch and advanced data modeling; Moodle platform customization and integration; DevOps practices with Docker; plus software architecture design, OOP principles, Agile methodologies, and web services (XML/REST).

LAICA - Laboratório de Pesquisa Allyson Amilcar Angelus
IT Researcher
July 2022 - March 2025 (2 years 9 months)
Natal, RN

Moodle integration and plugin development

Núcleo Avançado de Inovação Tecnológica (NAVI)
IT Researcher
July 2020 - December 2021 (1 year 6 months)
Natal, RN

Develops Electronic Health Record (EHR) to Rio Grande do Norte state

VOX Tecnologia da Informação Ltda
Software engineer
February 2013 - January 2014 (1 year)
João Pessoa, Paraiba, Brazil

1. Definition of the new software development methodology.
2. Management of software architecture, infrastructure and database teams.
3. Maintenance of REDESIM management software.
4. Correction of security flaws in the Zend Framework.

TSE - Tribunal Superior Eleitoral
Software Architect
March 2010 - February 2013 (3 years)

Brasília, DF

Software architect specializing in Java enterprise solutions, I defined architectural standards, software engineering practices, and scalable system design while providing expert consultancy on security projects. My technical expertise includes Java development with Spring framework and Hibernate ORM, application servers like Tomcat and JBoss, advanced data modeling across PostgreSQL and Oracle databases, object-oriented programming principles, web services integration, JavaScript frontend development with CSS styling, and comprehensive software development lifecycle management.

Optimedia Tecnologia em Informática Ltda

Assistant Project Manager

April 2009 - February 2010 (11 months)

Brasília, DF

Led team development of social networking software and distance e-learning platforms (including video conferencing features) at a startup, managing full-stack implementation from backend to frontend. My technical expertise encompassed PHP development with phpBB and WordPress CMS, Moodle e-learning platform customization, JavaScript frontend with CSS styling, backend development with SQL/MySQL databases and advanced data modeling, object-oriented programming principles, web services integration, Agile methodologies, and comprehensive web development practices for scalable online learning and social platforms.

FNDE - Fundo Nacional de Desenvolvimento da Educação

6 years 10 months

Software architect manager

September 2004 - February 2009 (4 years 6 months)

Brasília, DF

Led enterprise architecture transformation projects, defining and managing distributed Web Services and cluster environments to integrate legacy systems with modern agency projects—including a Moodle 1.3 implementation study. Improved Java legacy architecture (JasperReports, Struts 1, Hibernate 2, Oracle9i IAS) and maintained Delphi 5 VCL applications with custom components, while architecting progressive migrations from Delphi 5 to Delphi 2005 VCL and enabling rapid ASP.NET development. Designed productive new Java architectures supporting distributed processing (EJB3, JSF, clustering) and developed a Rich Internet Application (RIA) for electronic accountability integrated with protocol, financial, and budgetary control systems using ExtJS and JSON. Technical expertise included Java/Spring/

Hibernate on Tomcat/JBoss, Delphi development, SQL/PLSQL across Oracle/ PostgreSQL/MySQL, advanced data modeling, JavaScript/CSS frontend, XML web services, REST API development, OOP principles, Agile methodologies, Moodle platform evaluation, and full-stack web/software development.

Java Project Designer
February 2004 - September 2004 (8 months)
Brasília, Federal District, Brazil

1. Construction of the corporate business class model in Java to strengthen and mature the "Software Development Methodology (MDS)" due to the gain with reuse for reducing the estimated time for building new systems.

Delphi Developer
May 2002 - February 2004 (1 year 10 months)
Brasília, Federal District, Brazil

1. Development of an integrated financial management, budgeting and banking integration system that integrated with SIAFI and several banks (BB, Caixa, Itaú, etc.), the integrations were by object-oriented API that abstracted the specificities of the institutions allowed creation of specialized implementations for each institution.
2. Maintenance of an educational project elaboration system used by city halls, schools, etc ... in an interface and process that resembled the income tax declaration software of the Federal Revenue.

AgênciaClick
Web Application Developer
May 2001 - May 2002 (1 year 1 month)
Brasília, DF

Developed institutional websites, intranets, extranets, content publishing tools, and legacy system integrations using ASP, along with JSP-Java e-Commerce solutions. Technical expertise included Java backend development, SQL queries across MySQL databases with advanced data modeling, JavaScript frontend programming with CSS styling, object-oriented programming principles, and comprehensive web/software development practices for scalable enterprise applications.

Zevallos Tecnologia em Informática
Web developer
November 1999 - May 2001 (1 year 7 months)
Brasília, DF

Developed institutional websites, intranets, virtual school platforms, and content publishing systems using ASP and Hipertools, while creating and enhancing Hipertools—a VB6-based productivity framework for rapid web application development. Performed comprehensive error resolution, performance optimization, new feature implementation, and custom component development within Hipertools. Technical expertise included SQL database operations with advanced data modeling, JavaScript frontend development with CSS styling, backend programming, object-oriented principles, and full-stack web/software development practices for scalable enterprise solutions.

Poliedro

3 years 10 months

Delphi Developer

October 1996 - November 1999 (3 years 2 months)

Brasília, DF

Led comprehensive product development for Electronic Document Management (EDM/DMS) systems, including workflow engines, report generation IDEs, application builders (Access/Delphi/VB-style), installers, OCR/ICR API integrators, batch scanning via Automatic Document Feeders (ADF), automatic OCR/ICR document indexing, and library/archival science solutions. Technical expertise encompassed Delphi and Visual Basic application development, C++ for performance-critical components, legacy database systems (Clipper, DBase, Paradox, SQL Server) with advanced SQL data modeling, ActiveX component integration, object-oriented programming principles, and full-cycle software development practices for enterprise document automation solutions.

Clipper Developer

February 1996 - October 1996 (9 months)

Brasília, Federal District, Brazil

1. Development of Clipper systems for typist teams to carry out large-scale data loading concurrently in an MS-DOS environment over the Novel Network.

Education

Universidade Federal do Rio Grande do Norte

Specialization in Health, Medical Informatics · (2015 - 2017)

Universidade do Tocantins

Bachelor of Technology - BTech, Computer Systems Analysis/ Analyst · (2008 - 2010)